

Intelligent Drone Platform for Target Oriented with Full Autonomy

Perceive – Understand – Decide – Act

An advanced autonomous aerial system that integrates multi-modal sensing, AI perception, intelligent decision making and robust navigation to detect, track and engage targets with maximum autonomy, safety and mission effectiveness.

KEY CAPABILITIES



ADVANCED
SENSING



AI DETECTION &
TRACKING



AUTONOMOUS
NAVIGATION



COLLISION
AVOIDANCE



TARGET LOCK &
ENGAGEMENT



REAL-TIME DATA
& CONNECTIVITY



AI POWERED
PERCEPTION



TARGET ORIENTED
AUTONOMY



SAFE & RELIABLE
OPERATIONS



GNSS DENIED
CAPABILITY



NETWORKED
& SCALABLE



MISSION STATUS

AUTO

TARGET LOCK

ACTIVATED

LINK STATUS

SECURE

AUTONOMOUS MISSION WORKFLOW



SENSE

Multi-modal
Data Acquisition



PERCEIVE

AI Detection &
Scene Understanding



DECIDE

Mission Planning &
Target Evaluation



NAVIGATE

Autonomous Navigation
(GNSS / GNSS-Denied)



LOCK

Target Lock
(AI / Manual)



ENGAGE

Track & Act
with Precision



REPORT

Data Processing &
Mission Feedback

BUILT FOR COMPLEX MISSIONS

- Intelligence, Surveillance & Reconnaissance (ISR)
- Critical Infrastructure Monitoring
- Search, Detection & Tracking
- Maritime & Border Surveillance
- Disaster Response & Assessment

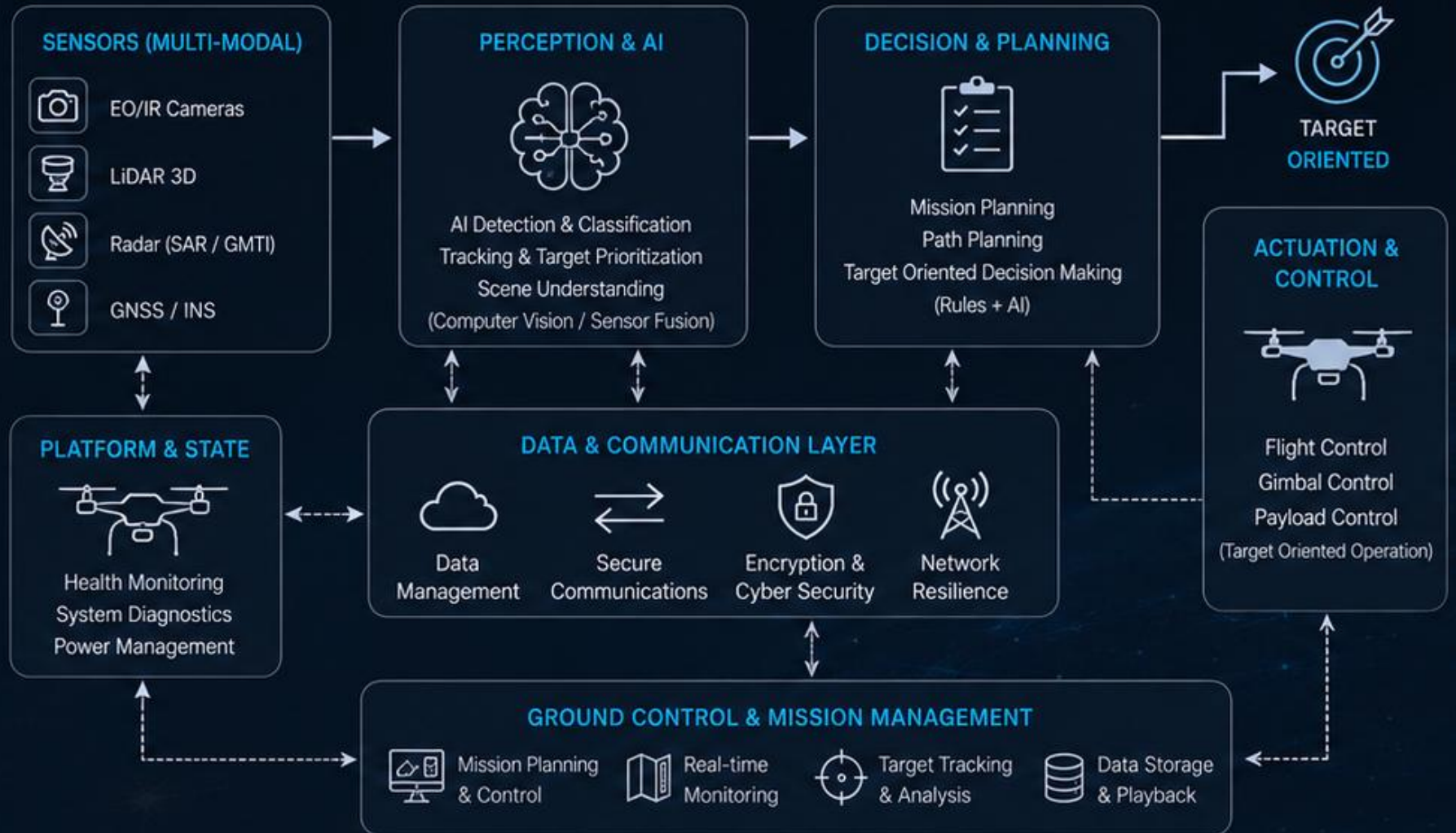
System Architecture Overview

MODULAR – SCALABLE – TARGET ORIENTED

Our architecture integrates multi-modal sensors, onboard AI perception, intelligent decision making and autonomous navigation to detect, track and engage targets with maximum autonomy, safety and mission effectiveness.

- ✓ **Target Oriented:** Detect, track and prioritize relevant targets.
- ✓ **Full Autonomy:** Sense, perceive, decide and act.
- ✓ **Robust & Secure:** Resilient in contested environments.
- ✓ **Mission Ready:** Designed for complex and critical missions.

ARCHITECTURE DIAGRAM



KEY ARCHITECTURE BENEFITS



TARGET ORIENTED

Focus on detecting, tracking and engaging relevant targets.



AI POWERED

Advanced AI for perception, classification and decisions.



FULL AUTONOMY

Autonomous navigation and mission execution.



SECURE & RESILIENT

Secure communications and resilient operations.



SCALABLE

Modular architecture for various missions.



MISSION EFFECTIVE

Maximizing effectiveness in complex environments.

System Integration & Interoperability

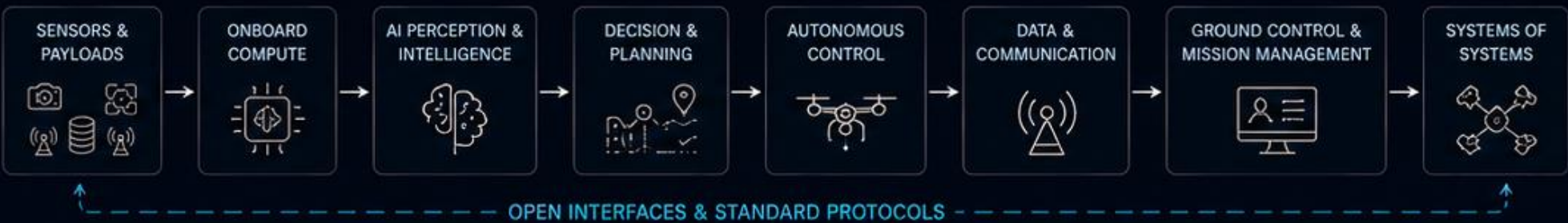
One System. Many Platforms.
Seamless Integration. Mission Ready.

Open, modular and interoperable architecture that connects sensors, platforms, payloads, networks and command systems to deliver superior situational awareness and mission effectiveness across domains.

KEY CAPABILITIES

- Modular & Open Architecture**
Plug-and-play components with well-defined interfaces and APIs.
- Cross-Platform Integration**
Air, ground, sea, space and personnel systems interoperability.
- Payload Agnostic**
Support for EO/IR, LiDAR, Radar, SIGINT, EW, kinetic and non-kinetic.
- Data & Command Interoperability**
Common data models, schemas and command protocols.
- Scalable & Future Ready**
Easily extendable to new sensors, platforms and mission systems.
- Standards Compliant**
Aligned with open standards and industry best practices.

SYSTEM INTEGRATION OVERVIEW



MULTI-DOMAIN PLATFORMS



DATA & COMMAND INTEROPERABILITY



PAYLOAD & SENSOR INTEGRATION



SOFTWARE INTEGRATION LAYER



PLATFORM & PAYLOAD MANAGEMENT



SECURITY & TRUST



BUILT ON PROVEN TECHNOLOGIES



VALIDATION & TESTING



CERTIFICATION & ASSURANCE



INTEROPERABILITY PRINCIPLES

- Open Standards First**
Use of open, modular, well-documented standards and protocols.
- Loose Coupling**
Decoupled services communicate via interfaces and messages.
- Data Centric**
Shared situational understanding through common data models.
- Security by Design**
Secure, trusted and resilient interoperable operations.
- Domain Agnostic**
Operate across domains and mission types seamlessly.

INTERFACE STANDARDS

- SPACE ROS**
(ROS 2 for Space)
- ROS 2**
(Robotic Middleware)
- DDS**
Data Distribution Service
- MAVLink**
Telemetry & Command
- gRPC**
Remote Procedure Calls
- MQTT**
IoT Messaging
- OpenAPI / REST**
Web Services
- STANAG / MISB**
Military Standards

INTEGRATED. INTEROPERABLE. INTELLIGENT. ONE ECOSYSTEM. ENDLESS POSSIBILITIES. MISSION SUCCESS.

DATA FLOW

COMMAND FLOW

INFORMATION FLOW

Data & Communication Layer

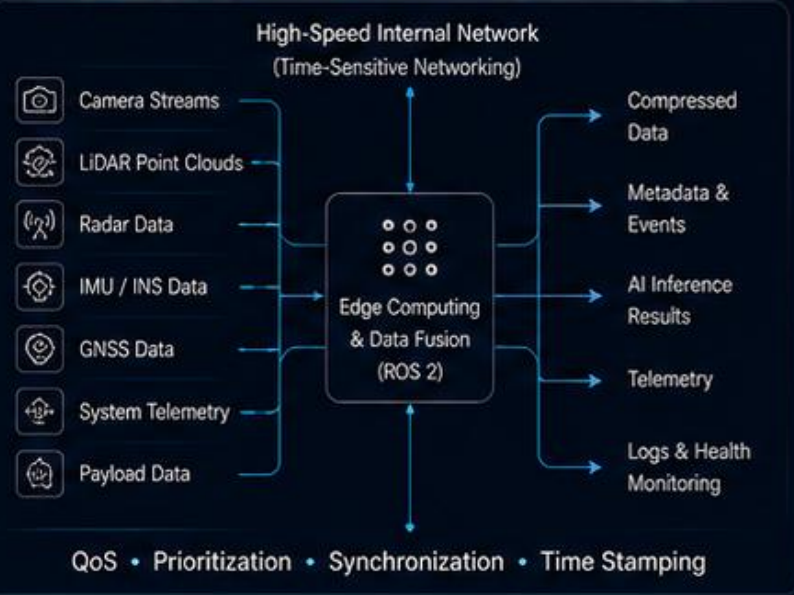
High-bandwidth, low-latency and secure communication infrastructure for robust data exchange, real-time telemetry and mission-critical operations in contested and GNSS denied environments.

- Reliable Connectivity**
Redundant and adaptive links ensure continuous operations.
- Low Latency**
Optimized for real-time control and data streaming.
- High Bandwidth**
Support for multi-sensor data, video streams and AI processing.
- Secure by Design**
End-to-end encryption, authentication and cyber resilience.
- Interoperable**
Built on open standards for seamless integration and scalability.

COMMUNICATION ARCHITECTURE OVERVIEW



ONBOARD DATA BUS & NETWORK



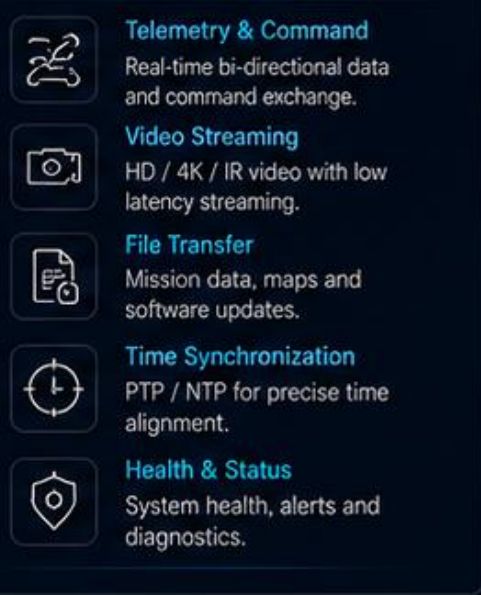
DATA LINKS



GROUND INFRASTRUCTURE



COMMUNICATION SERVICES



DATA FLOW



QUALITY OF SERVICE (QoS)



BUILT ON OPEN & PROVEN TECHNOLOGIES

SPACE ROS
Robotics Framework for Space

ROS 2
Real-time Robotic Middleware

MAVLink
Telemetry & Command Protocol

DDS
Data Distribution Service

PX4 Autopilot
Autopilot & Vehicle Integration

COMMUNICATION PROTOCOLS

MAVLink 2.0

RTSP / DDS

gRPC

WebRTC

MQTT

SFTP / FTPS

SECURITY & RESILIENCE

End-to-End Encryption

Authentication & Authorization

Intrusion Detection

Anti-Jamming & Anti-Spoofing

Redundant Links & Failover

SECURE. RELIABLE. RESILIENT. CONNECTING DATA. ENABLING AUTONOMY.

Open Standards | **Modular** | **Scalable** | **Interoperable** | **Future Ready**

Verification, Validation & Assurance

Trust the System.
Assure the Mission.

Rigorous verification, validation and assurance across the entire lifecycle to ensure safety, security, reliability and mission success in all conditions.

KEY CAPABILITIES



Lifecycle Assurance

End-to-end V&V from concept, design and development to operations and disposal.



Multi-Level Testing

MIL, SIL, HIL, PIL and Field testing across hardware, software and systems.



Standards & Compliance

Align with DO-178C, DO-254, DO-326A, ISO 26262, ISO 21448, STANAG, NIST and other applicable standards.



Safety & Reliability

System safety analysis, FMEA, FTA, HAZOP, fault injection and reliability growth.



Security Assurance

Threat modelling, penetration testing, code analysis, secure development lifecycle and supply chain security.



Continuous Assurance

Continuous integration, regression testing, monitoring and configuration management.

V&V LIFECYCLE



V&V METHODS & ACTIVITIES



ASSURANCE FRAMEWORK



TESTING STRATEGY & ENVIRONMENTS



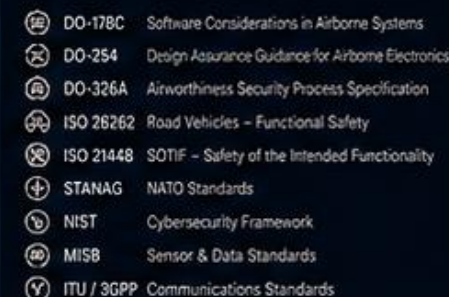
ASSURANCE EVIDENCE



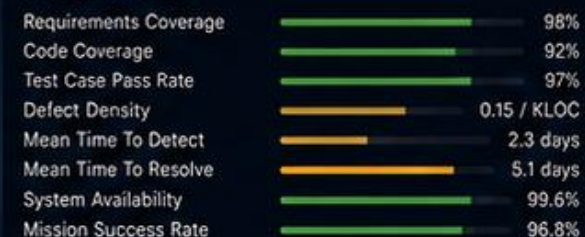
ASSURANCE DOMAINS



COMPLIANCE & STANDARDS



KPIs & METRICS



BUILT ON PROVEN TECHNOLOGIES



TOOLS & PLATFORMS



CERTIFICATION & ASSURANCE SUPPORT



VERIFIED DESIGN.



VALIDATED PERFORMANCE.



ASSURED OPERATION.



TRUSTED SYSTEMS.

ASSURANCE IS NOT A PHASE. IT'S A MINDSET.

AI Perception & Target Oriented

See • Understand • Describe • Decide

Advanced AI perception transforms multi-modal sensor data into actionable intelligence. It detects, classifies, tracks and understands scenes and targets, providing rich descriptions and context for autonomous decision making.

KEY CAPABILITIES

- AI Detection & Classification**
Detect and classify objects, vehicles, people and activity with high accuracy.
- Tracking & Target Prioritization**
Multi-target tracking with re-identification and dynamic priority scoring.
- Scene Understanding**
Understand environment, terrain, objects, relationships and activities.
- LLM Visual Description**
Generate natural language descriptions and summaries of scenes and targets.
- Anomaly & Threat Detection**
Detect unusual behavior and potential threats in real-time.

BUILT ON PROVEN TECHNOLOGIES

 **SPACE ROS**
Robotics Middleware for Space & Air

 **OpenCV**
Open Source Computer Vision

 **PyTorch**
Deep Learning Framework

 **TensorFlow**
Deep Learning Framework

 **MAVLINK**
Telemetry & Data Protocol

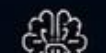
PERCEPTION QUALITIES

 **ACCURATE**

 **ROBUST**

 **REAL-TIME**

 **SCALABLE**

 **ADAPTIVE**

PERCEPTION PIPELINE



 **AI MODELS & FRAMEWORKS**
YOLOv8/11, Detectron2, DeepSORT, Transformer Models, SAM, CLIP, LLaVA, LLMs and more.



LLM POWERED VISUAL DESCRIPTION

Generates human-like descriptions and summaries from visual inputs. Enables natural interaction and better situational awareness.



"A white SUV is moving north on a dirt road near a building. Two individuals are standing next to a parked pickup truck. The area is surrounded by trees and low vegetation."

Sensor Layer, LLM Visual Descriptions

See. Understand. Describe.
Act with Context.

Multi-modal perception fused with
LLM-powered visual understanding to
generate rich, actionable, natural
language descriptions of the
operational environment.

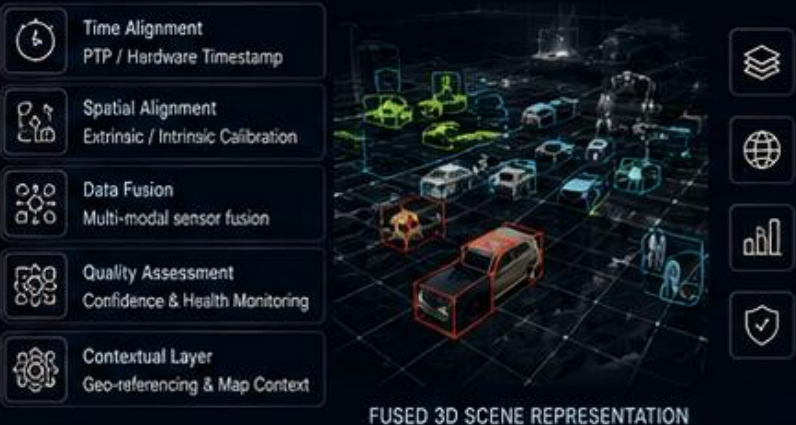
KEY CAPABILITIES

- Comprehensive Sensing**
EO/IR, LIDAR, Radar, SIGINT and environmental sensors.
- Multi-Modal Fusion**
Spatial & temporal alignment, calibration and data fusion.
- LLM Visual Understanding**
Generate detailed, contextual descriptions of scenes and events.
- Actionable Insights**
Convert descriptions into structured intelligence for decision making.
- Continuous Learning**
Feedback loop to improve models, prompts and performance.
- Explainable & Traceable**
Every description is linked to source data and confidence.

1. SENSOR LAYER (MULTI-MODAL)



2. DATA INGESTION & FUSION



3. LLM VISUAL DESCRIPTION PIPELINE



4. VISUAL INPUT EXAMPLES



5. LLM GENERATED DESCRIPTIONS (EXAMPLES)

- "A pickup truck is parked near a small building. Two individuals are standing beside it, appearing to load items into the bed."
Confidence: 0.91
- "Multiple tents and temporary shelters observed in an open area. Foot traffic detected between structures."
Confidence: 0.88
- "A boat is moving west along the river. Wake pattern suggests moderate speed. Possible smuggling route."
Confidence: 0.90
- "Heat signatures indicate a group of people gathered around a heat source during nighttime."
Confidence: 0.93

6. ACTIONABLE INTELLIGENCE OUTPUTS

- Mission Updates
Situational awareness summaries
- Threat Assessment
Risk scoring & prioritization
- Targeting Support
Identify & track objects of interest
- Decision Support
Contextual insights for commanders
- Reporting
Auto-generate mission reports

7. TECHNOLOGY STACK



8. ASSURANCE & GOVERNANCE



Decision & Planning (Target Oriented)

Intelligent decision making and mission planning focused on target objectives, situation awareness and safe execution in complex and dynamic environments.

-  **Mission Planning**
Define objectives, constraints, priorities and rules of engagement.
-  **Path Planning**
Generate optimal and safe routes considering terrain, threats and no-fly zones.
-  **Target Oriented Decision Making**
AI-driven decisions to select, track and engage priority targets.
-  **Re-planning**
Dynamic re-planning based on environment and mission changes.



MISSION PLANNING & PATH OPTIMIZATION



CONSTRAINTS CONSIDERED



TARGET ORIENTED DECISION ENGINE

-  **Target Prioritization**
Score and rank targets based on mission objectives, threat level and opportunity.
-  **Engagement Strategy**
Select best engagement method, approach vector and timing.
-  **Resource Allocation**
Optimize use of sensors, payload, energy and communication.
-  **Risk Assessment**
Evaluate risks and mission impact before execution.
-  **Go / No-Go Decision**
Validate mission feasibility and operational safety.

DIGITAL TWIN & SIMULATION



Validate plans and decisions in a high-fidelity digital twin environment.

- ✓ What-if Analysis
- ✓ Risk & Outcome Prediction
- ✓ Plan Validation
- ✓ Mission Rehearsal

INTEGRATED WITH

 **SPACE ROS**
Robotics Framework for Autonomy

 **MAVLink**
Telemetry & Command Protocol

 **DDS**
Data Distribution Service

 **FMI / FMU**
Model Exchange & Co-Simulation

 **Digital Twin**
Simulation & Validation

DEVELOPMENT & VALIDATION



CERTIFICATION & ASSURANCE



 **INTELLIGENT DECISIONS. SAFE MISSIONS. TARGET ORIENTED AUTONOMY.**

Ground Control & Mission Management

Command. Monitor. Understand.
Decide with Confidence.

Unified ground control station for mission planning, real-time monitoring, payload control and team collaboration.
Designed for autonomy at scale.



Mission Planning

Intuitive 2D/3D mission planner with waypoints, geofences, triggers and payload tasks.



Real-time Monitoring

Live telemetry, system status, sensor feeds and alerts in a single operational picture.



Payload Control

Control EO/IR, LiDAR, radar and effectors with real-time tasking and feedback.



Team Collaboration

Multi-operator roles, chat, annotations and shared situational awareness.



Data Management

Record, store, tag and replay mission data with intelligent search and export.



Alert & Notification

Configurable alerts, system health and mission event notifications.

GCS CAPABILITIES



PLAN

Plan & organize missions



MONITOR

Real-time ops monitoring



CONTROL

Vehicle & payload control



ANALYZE

Data analysis & visualization



COLLABORATE

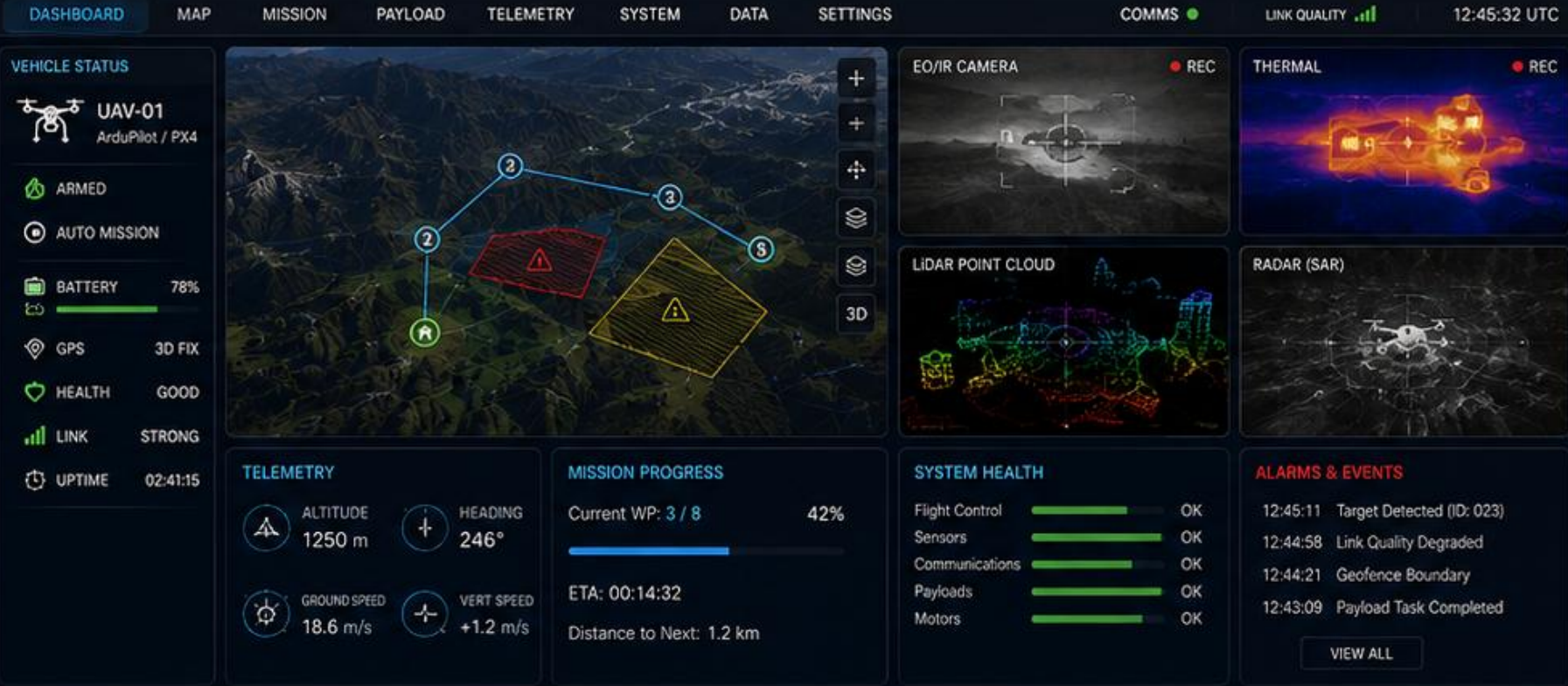
Team ops & communication



REPORT

Reports & mission debrief

GROUND CONTROL STATION (GCS) INTERFACE



PAYLOAD MANAGEMENT



EO/IR



LiDAR



Radar



SIGINT



Comms Relay



Effectors

DATA MANAGEMENT



Record



Store



Tag



Search



Replay



Export

VIDEO STREAMING



Low Latency



Adaptive Bitrate



Multi-Stream Support



H.264 / H.265 / H.265+

TIME SYNCHRONIZATION



PTP (IEEE 1588)

Sub-microsecond sync



NTP

Network time sync

BUILT ON OPEN & PROVEN TECHNOLOGIES



SPACE ROS
Robotics Framework for space



ROS 2
Next-gen robotic middleware



DDS
Data Distribution Service



MAVLink
Telemetry & Command Protocol



Computer Vision



Qt
Cross-Platform UI Framework

COMMUNICATION PROTOCOLS



DDS
Real-time Data



MAVLink 2
Telemetry / CMD



gRPC
Remote Services



WebRTC
Video / Data



MQTT
IoT Messaging



SFTP / FTPS
Secure Transfer

SECURITY & RESILIENCE



End-to-End Encryption



Auth & Access Control



Anti-Jamming Techniques



Link Redundancy & Failover



Data Integrity & Signing

MISSION MANAGEMENT



Mission Library
Create, save and reuse mission templates and packages.



Geofencing
Define no-fly, inclusion and exclusion zones.



Triggers & Events
Time, distance, condition and custom event triggers.



Auto / Manual Switch
Seamless transition between autonomous and manual control.



Return to Base
Smart RTL with dynamic recovery and fail-safe.

TEAM & COLLABORATION



Multi-Operator Console
Role-based access and simultaneous operations.



Voice & Chat
Secure voice, text and mission coordination.



Annotations & Markers
Share annotations, POIs and mission notes.



Live Sync
Real-time sync across teams and devices.



ONE CONSOLE. COMPLETE CONTROL. MAXIMUM AWARENESS. MISSION SUCCESS.



MODULAR



SCALABLE



INTEROPERABLE



SECURE



FUTURE READY

Autonomous Mission Workflow

Perceive. Decide. Act. Adapt.
Fully Autonomous. Always Aware.

End-to-end autonomous mission execution with closed-loop decision making, real-time adaptation and continuous verification in GNSS denied and contested environments.

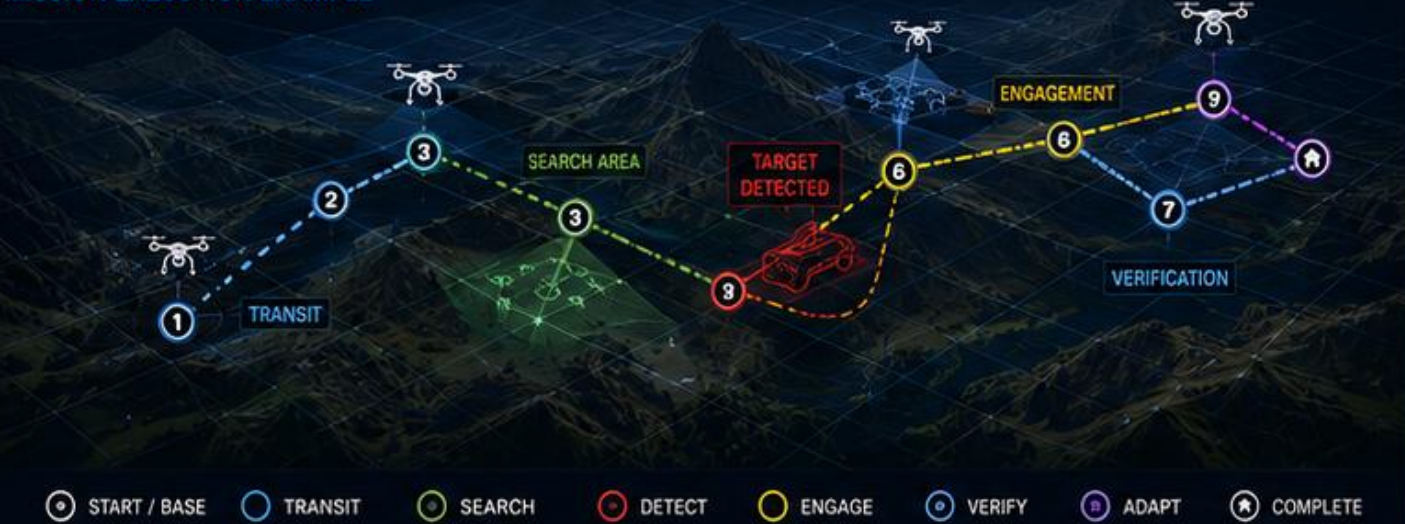
KEY CAPABILITIES

- End-to-End Autonomy**
From mission start to completion with minimal human intervention.
- Closed-Loop Adaptation**
Continuously sense, decide, act and learn in dynamic environments.
- Robust & Resilient**
Operates in GNSS denied, contested and degraded conditions.
- Mission Assurance**
Continuous health monitoring, verification and automatic re-planning.
- Objective Focused**
Target oriented execution with intelligent prioritization and verification.

AUTONOMOUS MISSION LIFECYCLE



MISSION EXECUTION EXAMPLE



AUTONOMOUS DECISION FACTORS



AUTONOMOUS WORKFLOW PRINCIPLES



FAIL-SAFE & CONTINGENCY



MISSION OUTCOMES

- ✓ Primary Objectives Achieved
- ✓ Targets Engaged / Neutralized
- ✓ Data Collected & Delivered
- ✓ System Safe & Recovered
- ✓ Mission Success

WORKFLOW BENEFITS

- High Autonomy, Low Workload
- Faster Decisions, Better Outcomes
- Operates in Complex Environments
- Scalable, Reliable, and Repeatable

BUILT ON PROVEN TECHNOLOGIES

SPACE ROS
Robotics Framework for Space

ROS 2
Real-time Robotic Middleware

DDS
Data Distribution Service

MAVLink
Telemetry & Command Protocol

PX4
Autopilot Autonomy Stack

Computer Vision

PyTorch
Deep Learning Framework

TensorFlow
Deep Learning Framework

VALIDATION & TESTING



CERTIFICATION & ASSURANCE



Automatic Detection & Tracking Workflow

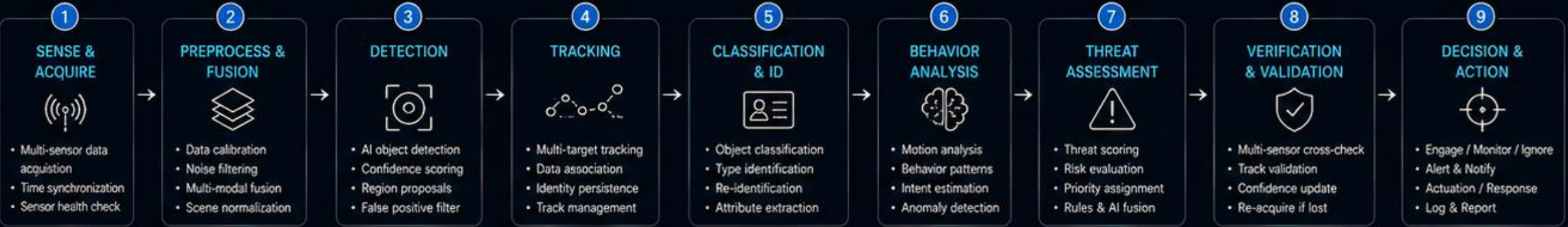
Detect. Track. Classify.
Understand. Act.

Fully autonomous target detection and tracking pipeline with continuous verification and closed-loop adaptation.

KEY CAPABILITIES

- Automatic Detection**
AI models detect objects of interest in real-time.
- Multi-Target Tracking**
Track multiple targets across frames with identity persistence.
- Classification & ID**
Classify targets and identify types (vehicle, person, vessel, etc.).
- Behavior Analysis**
Understand patterns, intent and potential threat level.
- Continuous Verification**
Verify detections with multi-sensor fusion and re-identification.
- Adaptive & Resilient**
Adapt to environment changes and maintain tracking robustness.

AUTOMATIC DETECTION & TRACKING PIPELINE



MULTI-MODAL SENSOR INPUTS

- EO/IR Camera**
High-resolution day & night imagery
- LiDAR**
3D point cloud and object structure
- Radar (SAR / FMCW)**
All-weather detection and velocity
- GNSS / INS**
Position, velocity, orientation
- Other Sensors**
Acoustic, RF, Chemical, Environmental

AI PERCEPTION & SENSOR FUSION



AI MODELS: YOLOv8 / YOLO-NAS • DeepSORT / ByteTrack • PointPillars • EfficientNet • Transformers • LLM/VLM for Description

TRACKING OUTPUTS

- Real-time Tracks**
ID, position, velocity, heading, confidence
- Track History**
Complete timeline and movement history
- Target Attributes**
Type, size, color, speed, direction, behavior
- Live Alerts**
Priority targets and threat notifications
- Mission Integration**
Data to GCS, weapons or other systems
- Data Logging**
Store for review, training and audit

TYPICAL USE CASES

- Ground Vehicles Detection & Tracking
- Maritime Vessels Detection & Tracking
- Personnel Detection & Tracking
- Fixed / Static Assets Monitoring
- Anomaly Detection
- Convoy Tracking
- Search & Rescue Operations
- Border / Area Surveillance

AI ENRICHMENT & LLM DESCRIPTION

- Natural Language Visual Description
- Scene Summarization
- Intent & Context Understanding
- Report Generation Auto / On Demand

OUTPUT INTEGRATION

- MAVLink Telemetry
- DDS Data Bus
- ROS 2 Topics
- REST API / WebSocket

BUILT ON OPEN & PROVEN TECHNOLOGIES

- SPACE ROS
Robotics Framework for Space
- ROS 2
Real-time Robotic Middleware
- DDS
Data Distribution Service
- MAVLink
Telemetry & Command Protocol
- OpenCV
Computer Vision
- PyTorch
Deep Learning Framework
- TensorFlow
Deep Learning Framework

VALIDATION & TESTING

- MIL
(Model-in-the-Loop)
- SIL
(Software-in-the-Loop)
- HIL
(Hardware-in-the-Loop)
- VIL
(Vehicle-in-the-Loop)

CERTIFICATION & ASSURANCE

- DO-178C
Software Integrity
- DO-254
Hardware Integrity
- DO-326A
Cybersecurity
- Safety & Risk Management
- Traceability & Compliance



AUTOMATIC DETECTION. INTELLIGENT TRACKING. CONTEXT AWARE. MISSION FOCUSED. ENABLING AUTONOMY. DELIVERING SUPERIOR SITUATIONAL AWARENESS.

Target Intelligence & Threat Assessment

Understand the Intent.
Assess the Risk. Act with Advantage.

Advanced AI models and multi-source intelligence fusion deliver deep understanding of targets and threats, enabling autonomous systems to prioritize, adapt and respond effectively in contested environments.

KEY CAPABILITIES

- Target Identification**
Detect and identify targets of interest across multiple domains and modalities.
- Threat Classification**
Classify threats by type, capability and intent using AI and rule-based logic.
- Intent & Behavior Analysis**
Analyze patterns and behaviors to infer intent and potential actions.
- Risk Assessment**
Evaluate threat level and mission impact to support decision making.
- Target Prioritization**
Rank targets based on mission value, threat level and opportunity window.
- Context Awareness**
Fuse environment, mission and operational context for better decisions.
- Continuous Learning**
Improve models and rules with feedback, outcomes and operator input.

BUILT ON PROVEN TECHNOLOGIES

- SPACE ROS**
Robotics Framework for Space
- ROS 2**
Real-time Robotic Middleware
- DDS**
Data Distribution Service
- MAVLink**
Telemetry & Command Protocol
- OpenCV**
Computer Vision Library
- PyTorch**
Deep Learning Framework
- TensorFlow**
Deep Learning Framework

DATA MANAGEMENT

- Data Lake Storage**
- Metadata Catalog**
- Versioning & Lineage**
- Access Control & Audit**
- Data Quality Monitoring**

VALIDATION & TESTING

- MIL**
(Model-in-the-Loop)
- SIL**
(Software-in-the-Loop)
- HIL**
(Hardware-in-the-Loop)
- VIL**
(Vehicle-in-the-Loop)

CERTIFICATION & ASSURANCE

- DO-178C**
Software Integrity
- DO-254**
Hardware Integrity
- DO-326A**
Cybersecurity
- Safety & Risk Management**
- Traceability & Compliance**

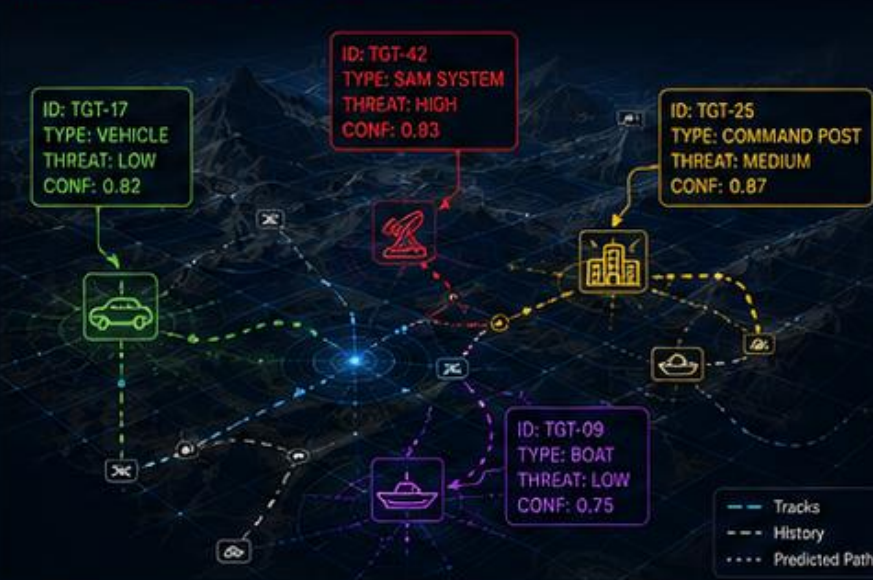
TARGET INTELLIGENCE PIPELINE



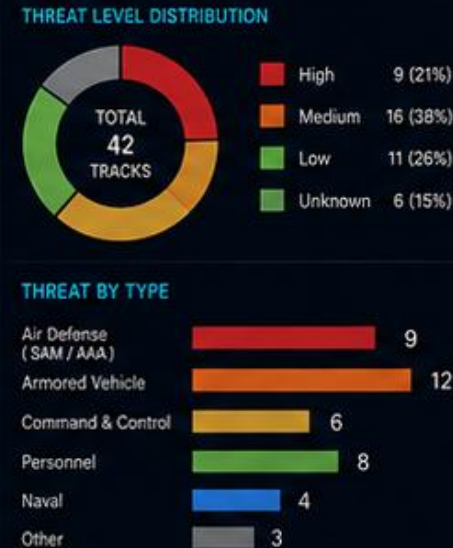
MULTI-SOURCE INTELLIGENCE INPUTS

- EO/IR Sensors**
Day/Night Imagery
- LiDAR Point Cloud**
3D Structure
- Radar (SAR / FMCW)**
Moving & Static Targets
- SIGINT / COMINT**
Signals & Communications
- HUMINT / OSINT**
Reports & Social Media
- GNSS / INS**
Position & Movement
- Environmental Data**
Weather, Terrain, EM

INTELLIGENCE FUSION & SITUATIONAL PICTURE



THREAT ASSESSMENT DASHBOARD



PRIORITIZED TARGET LIST

PRIORITY	TARGET	THREAT	CONFIDENCE
1	TGT-42 SAM System	HIGH	0.93
2	TGT-25 Command Post	MEDIUM	0.87
3	TGT-17 Armor Vehicle	LOW	0.82
4	TGT-09 Fast Boat	LOW	0.75
5	TGT-31 Radar Unit	MEDIUM	0.71

[VIEW ALL TARGETS](#)

AI MODELS & ANALYTICS

- YOLOv8 / YOLO-NAS**
Detection
- DeepSORT / ByteTrack**
Tracking
- EfficientNet / ResNet**
Classification
- Transformers**
Behavior Analysis
- Graph Neural Networks**
Relationship Modeling
- LLM / VLM**
Context & Description

THREAT INDICATORS

- Weapons Detected**
- Emission Detected**
- Convoy Movement**
- Fortification Activity**
- Unusual Behavior**
- No Comms / Jamming**

OUTPUTS & ACTIONABLE INSIGHTS

- Threat Alerts**
- Target Reports**
- Engagement Advice**
- Mission Impact**
- ROE Guidance**
- Intel Dissemination**

INTELLIGENT TARGETING. ACCURATE ASSESSMENT. INFORMED DECISIONS. MISSION ADVANTAGE. DATA DRIVEN. AI POWERED. HUMAN TRUSTED.

Repository Structure

Modular. Scalable. Maintainable. Mission Ready.

A unified, multi-package repository structure that supports all system layers, enables independent development and testing, and ensures traceability from code to mission.

DESIGN PRINCIPLES

-  **Separation of Concerns**
Clear boundaries between layers and components.
-  **Reusability**
Common libraries and interfaces across modules.
-  **Scalability**
Easily extendable to new payloads, platforms and missions.
-  **Testability**
Unit, integration and system testing at every layer.
-  **Traceability**
Requirements → Code → Tests → Deployment → Operations.
-  **Standards First**
Follow ROS 2, Space ROS, PX4, MAVLink and industry best practices.

REPOSITORY TREE

	Description
dsimetric-autonomy/	
01_docs/	System documentation, diagrams, and specifications
02_requirements/	System, software and hardware requirements
03_architecture/	Architecture models, interfaces and design docs
04_common/	Shared libraries, interfaces, message definitions
05_perception/	Sensor drivers, perception pipelines, detection & tracking
06_localization/	State estimation, sensor fusion, localization modules
07_mapping/	3D mapping, terrain analysis, environment models
08_planning/	Mission planning, path planning, task allocation
09_control/	Flight control, gimbal control, payload control
10_payload/	Payload drivers, payload apps, data processing
11_communication/	Telecom modules, protocols, data links, security
12_ground_control/	GCS apps, dashboards, mission management tools
13_simulation/	SIL/HIL environments, Gazebo/Isaac/Unreal setups
14_tests/	Unit, integration, system and regression tests
15_deployment/	Build, packaging, CI/CD, deployment manifests
16_tools/	Developer tools, scripts, code generators, utilities
17_configs/	Configuration files, parameters, YAML/JSON
18_data/	Datasets, maps, bag files, models (sample/reference)
19_logs/	Runtime logs, telemetry and diagnostics (gitignored)
.github/	CI/CD workflows, issue templates, codeowners
.gitignore	Git ignore rules
CHANGELOG.md	Change history and release notes
LICENSE	License information
README.md	Overview, setup, build and contribution guide

LAYER MAPPING

04_common/	→ Shared Foundation
05_perception/	→ Sensor Layer
06_localization/	→ Localization Layer
07_mapping/	→ Mapping Layer
08_planning/	→ Planning Layer
09_control/	→ Control Layer
10_payload/	→ Payload Layer
11_communication/	→ Communication Layer
12_ground_control/	→ GCS & Mission Layer
13_simulation/	→ Simulation & Test Layer
14_tests/	→ Verification Layer
15_deployment/	→ Deployment Layer
17_configs/	→ Configuration Layer
18_data/	→ Data Layer

CODE ORGANIZATION STANDARDS

- ✓ One package per feature/module
- ✓ ROS 2 package structure
- ✓ Meaningful naming conventions
- ✓ Interface definitions in /msg, /srv, /action
- ✓ Parameters in YAML
- ✓ Launch files in /launch
- ✓ Tests in /test
- ✓ Documentation in /docs
- ✓ CI/CD for every package
- ✓ Semantic versioning

TECH STACK

	SPACE ROS 2	Robotics Framework
	PX4	Flight Control
	MAVLink	Telemetry & Command
	DDS	Data Distribution Service
	C++/Python	Core Languages
	Git	Version Control
	Docker	Containerization
	GitHub Actions	CI / CD Pipeline

PACKAGE STRUCTURE (EXAMPLE)

08_planning/	
mission_planner/	
include/	
mission_planner/	
planner.hpp	
src/	
planner.cpp	
launch/	
mission_planner.launch.py	
config/	
planner_params.yaml	
msg/	
Mission.msg	
srv/	
PlanPath.srv	
package.xml	
CMakeLists.txt	
test/	
test_planner.cpp	

```
planner_params.yaml
planner:
  max_velocity: 15.0
  max_acceleration: 5.0
  safety_distance: 10.0
  replan_rate_hz: 2.0
costmap:
  resolution: 0.5
  inflation_radius: 3.0
  unknown_cost: 100
```

```
CMakeLists.txt (snippet)
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)

add_library(${PROJECT_NAME}_node src/planner.cpp)

ament_target_dependencies(${PROJECT_NAME}_node
  rclcpp nav_msgs geometry_msgs)

install(TARGETS ${PROJECT_NAME}_node
  DESTINATION lib/${PROJECT_NAME})
```

WORKFLOW



CI/CD PIPELINE (EXAMPLE)



BRANCHING STRATEGY



QUICK COMMANDS

```
# Clone repository
git clone https://github.com/dsimetric/dsimetric-autonomy.git

# Build workspace
colcon build --symlink-install

# Source workspace
source install/setup.bash

# Run tests
colcon test

# Launch example
ros2 launch 08_planning mission_planner.launch.py

# Check package
ros2 pkg list | grep mission_planner
```


LOCKING TARGET MECHANISM – FIRST MODULE

User Selects Target → System Locks → Window Tracklets → Continuous Track

